SG No. 12 C_{2h}^3 $C2/m$ (b-axis setting) [monoclinic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$
* Wyckoff site: `2a`, site symmetry: `2/m`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$[0,\ 0,\ 0]$	`[1,-2,-3,4]`

Table 2: Wyckoff bond: `2b@2a`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0,\ Y,\ 0]$	$[0,\ 0,\ 0]$	`[1,2,-3,-4]`

Table 3: Wyckoff bond: `4c@2a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ 0]$	`[1,-3]`
2	$[-X,\ Y,\ -Z]$	$[0,\ 0,\ 0]$	`[2,-4]`

* Wyckoff site: `2b`, site symmetry: `2/m`

Table 4: Wyckoff bond: `2a@2b`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,-2,-3,4]`

Table 5: Wyckoff bond: `2b@2b`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0,\ Y,\ 0]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,-3,-4]`

Table 6: Wyckoff bond: `4c@2b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,-3]`
2	$[-X,\ Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,-4]`

* Wyckoff site: `2c`, site symmetry: `2/m`

Table 7: Wyckoff bond: `2a@2c`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-2,-3,4]`

Table 8: Wyckoff bond: `2b@2c`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,-3,-4]`

Table 9: Wyckoff bond: `4c@2c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-4]`

* Wyckoff site: `2d`, site symmetry: `2/m`

Table 10: Wyckoff bond: `2a@2d`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-3,4]`

Table 11: Wyckoff bond: `2b@2d`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,-3,-4]`

Table 12: Wyckoff bond: `4c@2d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-4]`

* Wyckoff site: `4e`, site symmetry: `-1`

Table 13: Wyckoff bond: `4a@4e`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[2,-4]`

* Wyckoff site: `4f`, site symmetry: `-1`

Table 14: Wyckoff bond: `4a@4f`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,-4]`

* Wyckoff site: `4g`, site symmetry: `2`

Table 15: Wyckoff bond: `4a@4g`, bond symmetry: `2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, y, 0]$	`[1,-2]`
2	$[-X, 0, -Z]$	$[0, -y, 0]$	`[3,-4]`

Table 16: Wyckoff bond: `4b@4g`, bond symmetry: `2`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[0, y, 0]$	`[1,2]`
2	$[0, -Y, 0]$	$[0, -y, 0]$	`[3,4]`

Table 17: Wyckoff bond: `8c@4g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, 0]$	`[1]`
2	$[-X, Y, -Z]$	$[0, y, 0]$	`[2]`
3	$[-X, -Y, -Z]$	$[0, -y, 0]$	`[3]`
4	$[X, -Y, Z]$	$[0, -y, 0]$	`[4]`

* Wyckoff site: `4h`, site symmetry: `2`

Table 18: Wyckoff bond: `4a@4h`, bond symmetry: `2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, y, \frac{1}{2}\right]$	`[1,-2]`
2	$[-X, 0, -Z]$	$\left[0, -y, \frac{1}{2}\right]$	`[3,-4]`

Table 19: Wyckoff bond: `4b@4h`, bond symmetry: `2`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, y, \frac{1}{2}\right]$	`[1,2]`
2	$[0, -Y, 0]$	$\left[0, -y, \frac{1}{2}\right]$	`[3,4]`

Table 20: Wyckoff bond: `8c@4h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, y, \frac{1}{2}\right]$	`[1]`
2	$[-X, Y, -Z]$	$\left[0, y, \frac{1}{2}\right]$	`[2]`
3	$[-X, -Y, -Z]$	$\left[0, -y, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, Z]$	$\left[0, -y, \frac{1}{2}\right]$	`[4]`

* Wyckoff site: `4i`, site symmetry: `m`

Table 21: Wyckoff bond: `4a@4i`, bond symmetry: `m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[x, 0, z]$	`[1,4]`
2	$[-X, 0, -Z]$	$[-x, 0, -z]$	`[2,3]`

Table 22: Wyckoff bond: `4b@4i`, bond symmetry: `m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[x, 0, z]$	`[1,-4]`
2	$[0, Y, 0]$	$[-x, 0, -z]$	`[2,-3]`

Table 23: Wyckoff bond: `8c@4i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, z]$	`[1]`
2	$[-X, Y, -Z]$	$[-x, 0, -z]$	`[2]`
3	$[-X, -Y, -Z]$	$[-x, 0, -z]$	`[3]`
4	$[X, -Y, Z]$	$[x, 0, z]$	`[4]`

* Wyckoff site: `8j`, site symmetry: `1`

Table 24: Wyckoff bond: `8a@8j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, Y, -Z]$	$[-x, y, -z]$	`[2]`
3	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[3]`
4	$[X, -Y, Z]$	$[x, -y, z]$	`[4]`